Collaboration opportunities

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icteam — Institute of Information and Communication Technologies, Electronics, and Applied Mathematics

UCLouvain — Université catholique de Louvain

Outline

1. What is the ICTEAM Institute at UCLouvain
2. Collaboration opportunities with the faculty
   - Early-stage collaborations
   - R&D collaborations
   - Strategic partnerships
3. Conclusions and next steps
4. Discussion

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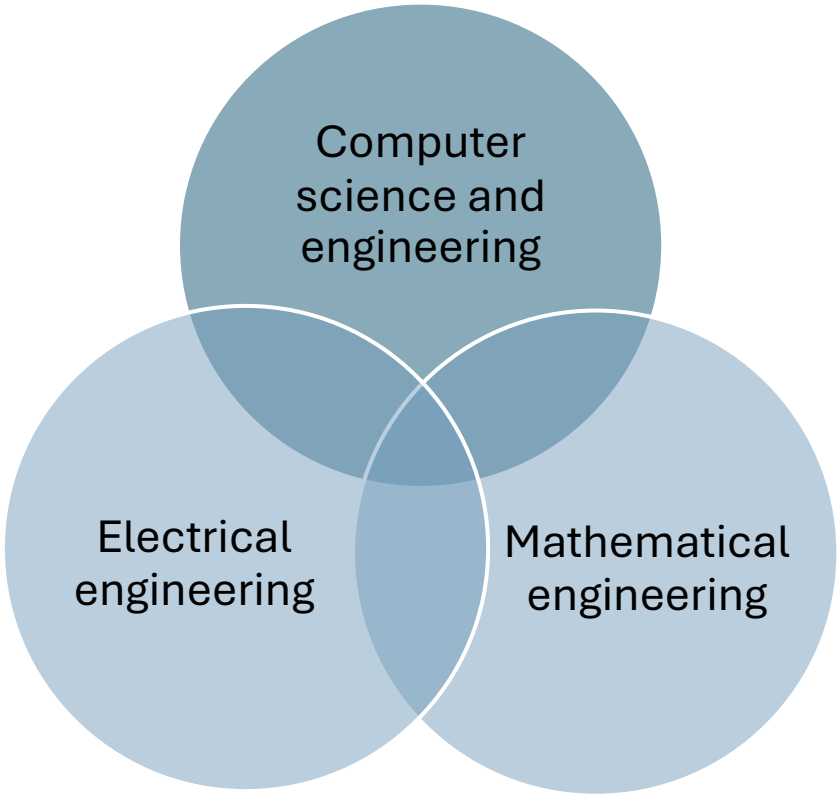
Electronics, and Applied Mathematics

http://uclouvain.be/en/icteam

The institute

An interdisciplinary research ecosystem



- 300 Researchers
- 50 Professor
- 9 Research themes
- 6 Application domains
- 3 Technology platforms

Areas of expertise

- Computer Systems & Networks
- Data Science and Artificial Intelligence
- Dynamical Systems, Control
- Communication Systems
- Algorithms & Optimization
- Electronics, Micro & Nano Technologies
- Software Engineering & Programming
- Biomedical Engineering & Data Analysis
- Cryptography & Information Security

Application domains

- Industry Processes & Manufacturing
- Space and Aerospace
- Health and Biomedical Technologies
- Culture, Education and Society
- Energy, Climate, Environment and Sustainability
- Security

Technology platforms

winfab
WALLONIA INFRASTRUCTURE NANO FABRICATION

- **W**allonia **IN**frastructure nano **FAB**rication
- 1,000 m² **cleanroom** infrastructure and a large high-tech equipment fleet (more than 50 machines)
- Offers access to a wide range of advanced techniques for thin-films deposition and etching, surface patterning, micromachining and back-end processes.

welcome
wallonia electronics and communications measurements

- The **W**allonia **EL**ectronics and **CO**mmunications **ME**asurements facility is a state-of-the-art technological platform
- Provides multidisciplinary tools in the field of **electrical and electromagnetic characterization**

OpenHub
Garage for innovators

- Innovation Hub
- Rapid prototyping development with MakiLab: VR, 3D printing, laser cutter, etc.
- Technical advice
- R&D collaborative projects
- Technical demonstration and events

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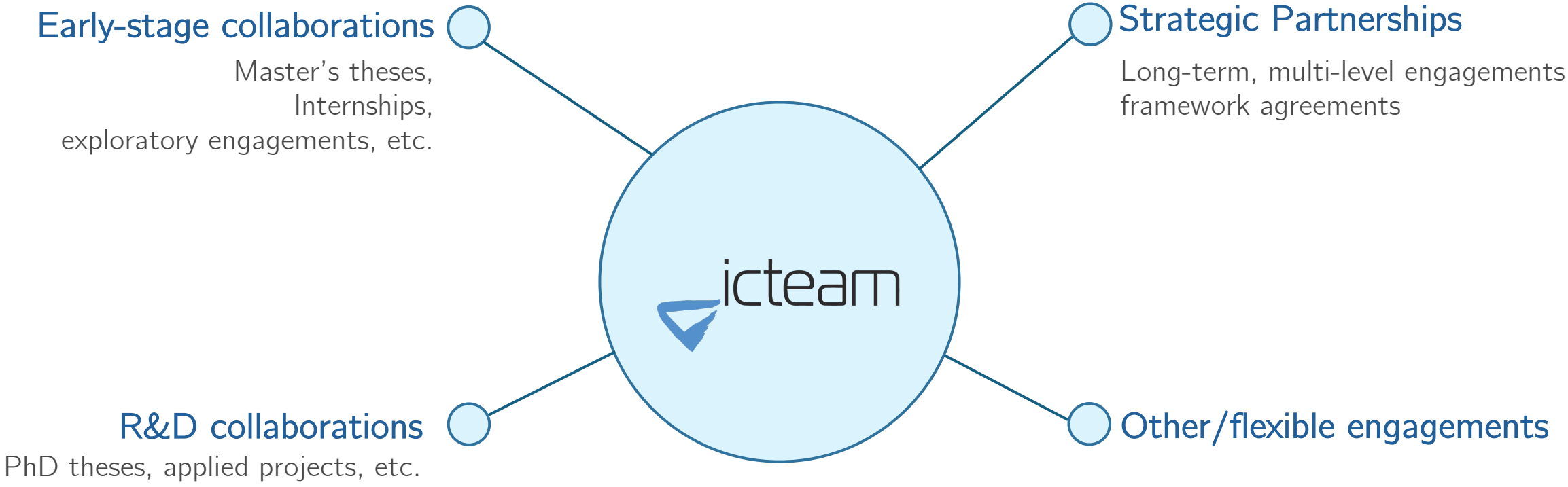
Collaboration opportunities

Opportunities for academic–industrial collaborations



Different levels of involvement, time-scales, and funding mechanisms

Early-stage collaborations

- **Research type:** exploratory, foundational (non strategic)
- **Format:** research is mainly conducted by Master's/PhD students, co-supervised by a faculty member and industry partner

- **Internships**
  - Typical duration: 9 weeks (during summer or during academic year)
  - Mainly supervised by company
  - Confidentiality agreement
  - Propose an internship topic at anytime, by contacting a faculty member

- **Master theses**
  - Typical duration: 1 year (student works at 50% time)
  - Supervision by professor + industry partner
  - Under IP agreement
  - The topic of the thesis is defined in collaboration with a faculty member. Topics must be posted around February of March of each year

- **Industry-inspired projects:** several options
  - Propose a topic for a course or semester project
  - Supervision by ICTEAM prof, ideally with regular contacts with company

Internships

- The company proposes a project related to the intern's studies and adapted to his/her level of training

- The intern spends physical time internally to the company

The project must:

- Be a complement to the scientific studies of the intern by discovering and learning professional practices.
- Expose the intern to the diverse technical, relational and managerial issues of the company;
- Focus on an industrial problem or technical project pertaining to the intern's specialization. The intern must actively participate in the project, side by side with the members of the company. No observational internships are allowed.
- Be non confidential
- Be non paid

Master theses

- Master theses range from technological and industrial projects (designed to introduce students to the engineering profession in industry) to applied or fundamental research projects (designed to introduce students to the engineering profession in research)
- Supervision is conducted in collaboration between professor and company representative
- Not paid
- Can potentially be combined with an itnernship

R&D collaborations

- **Research type:** Mid- to late-stage projects, focus on applied research and product development
- **Format:** research is mainly conducted by PhD students or postdocs
- **Intellectual property:** will be handled carefully based on needs through RTTO office

- Collaborative PhD project
  - Many options: Wallonia (e.g. Win4Doc), Brussels (e.g. Innoviris Applied PhD), EU (e.g. MSCA doctoral network, as beneficiary or internship host), etc.
  - Depending on type, initiative may come from industry or from university
  - 4 years typical for PhD
  - Ideally requires strong involvement (and/or co-supervision) by industry

- Postdoctoral Research Collaborations
  - E.g. postdoctoral fellowship MSCA (EU), 2–4 years

- Publicly Funded Applied Research Projects
  - E.g., Horizon Europe: 3 years, max 2,5M€, ARC (Action de Recherches Concertées) WBI

Strategic partnerships

Research type: Long-term, high-commitment collaborations with mutual strategic benefits

- Framework agreement and Call-Off contract for Industrial research (Igr/Post Doc/Prof consultancy)

- Sponsored Research Chairs:
  - 5 years typically
  - funding for permanent professor and research team
  - opportunity to suggest the institute focuses on a given area/topic

Other/flexible engagement

Research type: Lighter-touch or support-oriented collaborations

- **Consultancy** and expert analysis
- Coaching of employees (e.g. half-day or one-day trainings organized by ICTEAM)
- **Guest lectures**: industry professionals invited to attend and learn in cutting-edge courses
- **Sponsorships**: conferences, student event, competition/awards, hackathon
- **Company visits**: site visit organized in the context of some courses
- **Test your data**: use (anonymized) industrial data in course projects
- Suggest ideas for **Hackathons**

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Conclusion and next steps

Conclusions and next steps

- Collaboration modalities, the degree of company coordination, and grant preparation will vary depending on the situation and identified needs

- Contacts:
  - **ICTEAM coordinator: communication-icteam@uclouvain.be**
  - ICTEAM professors: contact directly, when you know/have identified them
    http://uclouvain.be/en/icteam
  - EPL for internship: epl-internships@uclouvain.be
  - WinFab: Christian Renaux christian.renaux@uclouvain.be
  - Welcome: Pascal Simon pascal.simon@uclouvain.be
  - OpenHub: Nicolas Bioul nicolas.bioul@uclouvain.be

Areas of expertise

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Computer Systems & Networks

Enhancing computational performance, resource sharing, and scalability across diverse environments, from personal devices to cloud infrastructures.

- **Cloud Computing & Distributed Systems** – Innovating in cloud-native architectures and large-scale distributed systems to enable next-generation applications.
- **Internet Measurement & Performance** – Advancing methodologies to assess, optimize, and secure network reliability, speed, and responsiveness.
- **Network Protocols and Standards** – Understanding and developing the protocols that govern communication across networks.
- **Large-Scale Systems Performance** – Improve the energy consumption, adaptability, efficiency and programmability of large-scale networked systems.
- **System Scalability** – Developing algorithms and frameworks to enhance large-scale system efficiency.
- **Embedded Systems & IoT** – Designing intelligent, resource-efficient embedded solutions that power the Internet of Things, enabling smart and connected environments.
- **Blockchain & Decentralized Systems** – Exploring the use of blockchain and related technologies for secure and transparent distributed systems.

Software Engineering and Programming Systems

Building reliable, efficient, and adaptive software solutions

- **System Scalability** – Architecting high-performance, resilient systems that adapt to dynamic network conditions and massive-scale computing environments.
- **Software Development & Maintenance** – Advancing methodologies for secure, modular, and maintainable software, from agile development to large-scale code evolution.
- **Programming Languages and Paradigms** – Exploring innovations in language design, compiler construction, and paradigm shifts that shape modern software development.
- **Program Quality Analysis and Testing** – Developing automated techniques for assessing software reliability, correctness, and security through static and dynamic analysis methods.
- **Computer Science Education Research** – Investigating pedagogical strategies and learning tools to enhance programming and software engineering education for diverse audiences.
- **Database Architecture** – Designing robust, scalable, and high-availability data systems that support efficient storage, retrieval, and management of structured and unstructured information across diverse applications.

Algorithms and Optimization

Solving complex, constrained, real-world problems across science, engineering, and society

- **Discrete Optimization Techniques** – Designing scalable, high-performance methods for combinatorial and integer optimization problems.
- **Continuous Optimization** – Solving real-time, constrained optimization problems central to control, decision-making, and resource allocation.
- **Algorithm Design & Efficiency** – Developing efficient algorithms with provable guarantees for large-scale and time-critical computations.
- **Constraint Satisfaction & Reasoning** – Solving high-dimensional decision problems under constraints with formal methods and heuristics in discrete spaces.

Contact:

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Data Science and Artificial Intelligence

Developing techniques in areas such as deep learning, natural language processing, computer vision, computational sensing and trustworthy AI

- **Machine Learning & Deep Learning** – Building adaptive models for prediction, classification, and generative modeling in complex domains.
- **Computer Vision & Image Processing** – Extracting structure and meaning from images and videos using advanced vision algorithms.
- **Natural Language Processing & Large Language Models** – Investigating the principles and applications of NLP and LLMs.
- **Language Contamination Detection** – Detecting bias, noise, and semantic drift in biomedical and scientific corpora.
- **Data Mining & Pattern Recognition** – Exploring the theoretical concepts for dealing with structured/unstructured high-dimensional data with unsupervised and supervised learning paradigms.
- **Dimensionality Reduction** – Representing and visualizing high-dimensional data with a limited number of features through selection and nonlinear embedding
- **AI Hardware & Implementation** – Co-designing hardware and software for efficient, real-time AI and ML deployment at scale.
- **Massive Data Systems** – Architecting scalable pipelines for processing massive, heterogeneous datasets.
- **Statistical Analysis & Inference** – Applying rigorous statistical frameworks to validate models, test hypotheses, and quantify uncertainty.

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Biomedical Engineering and Data Analysis

Designing medical devices, physiological modeling, and developing of computational methods for analyzing complex biomedical data

- **Database Architecture** – Designing robust, scalable, and high-availability data systems that support efficient storage, retrieval, and management of structured and unstructured information across diverse applications.
- **Healthcare Data Analysis** – Leveraging clinical and sensor-derived data for predictive modeling, risk assessment, and healthcare optimization.
- **Neuroscience** – Applying data-driven methods to neural systems, brain-computer interfaces, and physiological modeling.
- **Sensors & Embedded Systems** – Capturing and interpreting real-time biological signals through smart, wearable, and embedded platforms.

Communication Systems

Improving data transfer by enhancing telecommunications, antenna design, signal processing, data transmission, and channel understanding for reliability and efficiency

- **Communication Infrastructure** – Covering the foundations of telecommunications infrastructure, including wired and wireless networks, 5G/6G evolution, and next-generation data transmission.
- **Wireless & Radio Technologies** – Exploring radio frequency (RF) communication, mobile networking, and spectrum management to drive advancements in ubiquitous wireless connectivity.
- **Localization and Positioning** – Developing radar, lidar, and millimeter-wave sensing for environmental mapping, autonomous navigation, and real-time object detection.
- **Antennas and Propagation** – Advancing electromagnetic modeling and antenna technology, including improved ray-tracing tools, electromagnetic modeling of body area channels, and innovations in antenna and antenna array.
- **Signal Processing for Communication Systems** – Optimizing transmitters, receivers, and resource allocation for modern wired and wireless systems.

Control and Dynamical Systems

Designing control strategies that ensure system stability, performance, and adaptability in uncertain and dynamic environments.

- **Dynamical Systems Modeling** – Formulating mathematical representations of time-evolving systems across physical, biological, and engineered domains.
- **Control Theory** – Developing feedback and feedforward strategies to govern system behavior under constraints and uncertainty.
- **Hybrid and Nonlinear Systems** – Analyzing and controlling systems that combine continuous dynamics with discrete events and nonlinear interactions.
- **Robotics & Autonomy** – Applying control and planning techniques to enable adaptive, robust behavior in autonomous physical agents.

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Electronics, Micro & Nano Technologies

Developing advanced chips and hardware architectures that power modern electronic systems for sensing and computing.

- **Low-Power Circuit Design** – Engineering ultra-efficient analog and digital circuits for high-performance application
- **Internet of Things (IoT) Systems** – Designing integrated hardware systems for distributed, sensor-rich environments
- **Emerging Technologies** – Advancing integrated circuit technologies beyond traditional silicon
- **Nanomaterials for Smart Electronics** – Leveraging nanoparticles and ultra-thin films to enable tunable, responsive, and functional electronic systems.
- **Opto-electronics & Plasmonics** – Investigating light-matter interaction at the micro- and nanoscale for photonic and quantum
- **SOI Technology** – Enhancing device performance, energy efficiency and robustness to extreme environments through silicon-on-insulator architectures.
- **MEMS/NEMS** – Developing micro- and nano-electromechanical systems for sensing, actuation, and signal processing.
- **RF & Microwave Circuits and Devices** - Designing, modeling, fabricating and measuring active and passive microwave components and circuits (integrated and hybrid).
- **Electromechanical Characterization** – Investigating the mechanical, electrical, and thermal properties of substrates
- **RF Devices Characterization** – Performing on-wafer wideband characterization of advanced Silicon-on-Insulator MOSFETs
- **Environmental Impact Assessment and Mitigation** – Leveraging comprehensive life cycle assessments of electronic devices

Cryptography & Information Security

Bridging theory, systems, and hardware, this field provides the cryptographic and security foundations that underpin modern digital infrastructure—from secure messaging and e-voting to cloud computing and embedded systems.

- **Security Protocols** – Developing cryptographic frameworks for authentication, secure multiparty computation, verifiable voting, and zero-knowledge proofs.
- **Cryptographic Primitives** – Constructing the core algorithms for encryption, digital signatures, and secure key exchange using both classical and post-quantum techniques.
- **Implementation Security** – Analyzing and mitigating hardware and software vulnerabilities, including side-channel attacks and fault injection.
- **Privacy-Preserving Technologies** – Designing methods to safeguard sensitive data in storage, computation, and communication, across centralized and decentralized systems.
- **Software and Network Security** – Defending against malware, spam, and distributed attacks in both wired and wireless environments, including IoT and cyber-physical systems